1 Mark for Review

Which expression is equivalent to $(9x^2 + 9) - (2x^2)$?

A. 16

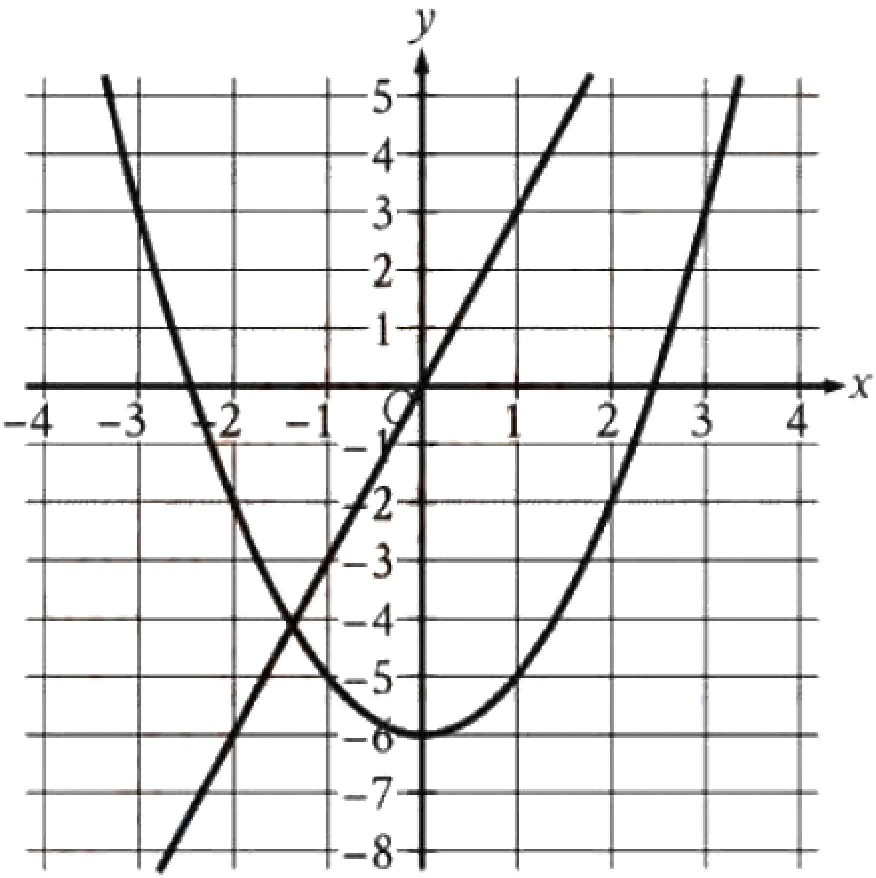
B. $11x^2 + 9$

C. $7x^2 - 9$

D. $7x^2 + 9$

2 Mark for Review

ABC



A. $y = -3x$
$y = x^2 + 6$

B. $y = -3x$
$y = x^2 - 6$

C. $y = 3x$
$y = x^2 + 6$

D. $y = 3x$
$y = x^2 - 6$

The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

3 Mark for Review

In triangle ABC, the measure of angle A is 31°, the measure of angle B is 90°, and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k?

A. 45
B. 59
C. 62
D. 118

4 Mark for Review

Number of cars	Maximum number of passengers and crew
3	111
6	216
10	356

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A) $35c - p = -6$

B) $35c - p = 6$

C) $35p - c = -6$

D) $35p - c = 6$

5 Mark for Review

ABC

$$g(x) = 5(16x - 17)$$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 5$ in the xy-plane?

(A) -90

(B) -85

(C) -22

(D) -17

6 Mark for Review

ABC

At the start of an experiment, the pressure inside a container was 18 pounds per square inch (psi). The pressure inside the container doubled every 29 minutes after the start of the experiment. Which equation represents this situation, where y is the pressure, in psi, inside the container x minutes after the start of the experiment?

A) $y = 18(2)^{\frac{x}{29}}$

B) $y = 18(29)^{2x}$

C) $y = 18(x + 29)^2$

D) $y = 18x^2 + 29$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

7 Mark for Review

$$|x - 8| + 72 = 88$$

What is the sum of the solutions to the given equation?

Answer Preview:

8 Mark for Review

ABC

What is the y-intercept of $y = -(8)^x - 38$ in the xy-plane?

A. $(0, -39)$

B. $(0, -37)$

C. $(0, -38)$

D. $(0, -46)$

Student-produced response directions

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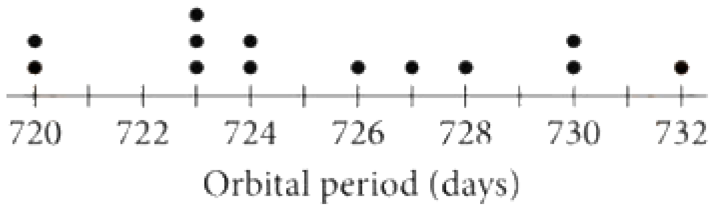
9 Mark for Review

$$x^2 + 9x + 6 = 0$$

One solution to the given equation can be written as $x = \frac{-9+\sqrt{k}}{2}$, where k is a constant. What is the value of k?

Answer Preview:

10 Mark for Review



A data set of the orbital periods, rounded to the nearest whole number of Earth days, for 13 of Jupiter's moons is represented in the dot plot. An additional moon with an orbital period of 621 days is added to the original data set to create a new data set of 14 orbital periods. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

(A) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

(B) The mean of the new data set is equal to the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(C) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.

(D) The mean of the new data set is less than the mean of the original data set, and the median of the new data set is equal to the median of the original data set.

Student-produced response directions

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11 Mark for Review

The function g is defined by $g(x) = 9x$. For what value of x is $g(x) = 63$?

Answer Preview:

12 Mark for Review

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x. The product of 14 and the fourth odd integer is at most 23 less than the sum of the first and third odd integers. Which inequality represents this situation?

A) $14(x+4) \geq 23 - (x + (x+3))$

B) $14(x+4) \leq x + (x+3) - 23$

C) $14(x+6) \geq 23 - (x + (x+4))$

D) $14(x+6) \leq x + (x+4) - 23$

13 Mark for Review

If $10+\frac{6(7-x)}{5}=\frac{4(7-x)}{3}$, which equation must also be true?

A) $7-x=148$

B) $7-x=-75$

C) $x-7=148$

D) $x-7=-75$

14 Mark for Review

The function f is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at $(1, 6)$ and passes through the points $(2, 34)$ and $(-1, 118)$. What is the value of $f(-2) - f(0)$?

Ⓐ 90

Ⓑ 146

Ⓒ 224

Ⓓ 258

Student-produced response directions

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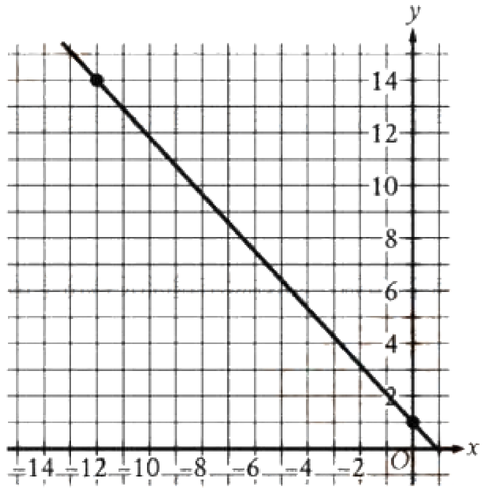
Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
	2/3 .6666	0.66

15 Mark for Review



The graph of line g is shown in the xy-plane. Line k is defined by $195x + py = w$, where p and w are constants. If line k is graphed in this xy-plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

Student-produced response directions

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16 Mark for Review

Triangles ABC and DEF are similar. Each side length of triangle ABC is 3 times the corresponding side length of triangle DEF. The area of triangle ABC is 170 square inches. What is the area, in square inches, of triangle DEF?

Answer Preview:

17 Mark for Review

After 60% of the original number of objects in a group were removed from the group, 300 objects remained in the group. How many objects were removed from the group?

Ⓐ 180

Ⓑ 450

Ⓒ 500

Ⓓ 750

Student-produced response directions

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18 Mark for Review

12, 12, 18, 24, 29

What is the median of the data set shown?

Answer Preview:

19 Mark for Review

$$y = x - c$$
$$y = -2(x - 12)^2$$

In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c?

A) 7

B) 11

C) $\frac{95}{8}$

D) 17

20 Mark for Review

ABC

The perimeter of an isosceles right triangle is $26 + 26\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

A) 13

B) $13\sqrt{2}$

C) 26

D) $26\sqrt{2}$

21. If n and k are numbers greater than 1 and $\sqrt[2]{n^5}$ is equivalent to $\sqrt[5]{k^4}$, for what value of a, n^{4a+1} is equal to k?

22 Mark for Review

An object's speed is increasing at a rate of 10.7 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

A) 0.4

B) 23.9

C) 150.4

D) 286.9